

Fig. 7.6 *Ascaris* male and female worm

Ascaris is a monogenic parasite and exhibits sexual dimorphism. **Ascariasis** is a disease caused by the intestinal endoparasite *Ascaris lumbricoides* commonly called the **round worms (Fig. 7.6)**. It is transmitted through ingestion of embryonated eggs through contaminated food and water. Children playing in contaminated soils are also prone to have a chance of transfer of eggs from hand to mouth. The symptoms of the disease are abdominal pain, vomiting, headache, anaemia, irritability and diarrhoea. A heavy infection can cause nutritional deficiency and severe abdominal pain and causes stunted growth in children. It may also cause enteritis, hepatitis and bronchitis.

Filariasis is caused by *Wuchereria bancrofti*, commonly called **filarial worm.** It is found in the **lymph vessels** and **lymph nodes** of man **(Fig. 7.7)**. *Wuchereria bancrofti* is sexually dimorphic, viviparous and digenic. The life cycle is completed in two hosts, man and the female *Culex* mosquito The female filarial worm gives rise to **juveniles** called **microfilariae larvae**. In the lymph glands, the juveniles develop into adults. The accumulation of the worms block the lymphatic system resulting in inflammation of the lymph nodes. In some cases, the obstruction of lymph vessels causes elephantiasis or filariasis of the **limbs**, **scrotum** and **mammary glands (Fig. 7.8)**.



Fig. 7.7 *Wuchereria bancrofti*



Fig. 7.8 Chronic inflammation of the lower limbs

7.2 Maintenance of personal and public hygiene

Hygiene is a set of practices performed to conserve good health. According to the World Health Organization (WHO), hygiene refers to "conditions and practices that help to maintain health and prevent the spread of diseases." **Personal hygiene** refers to maintaining one's body clean by bathing, washing hands, trimming fingernails, wearing clean clothes and also includes attention to keeping surfaces in the home and workplace, including toilets, bathroom facilities, clean and pathogen-free.

Our public places teem with infection, contamination and germs. It seems that every surface we touch and the air we breathe are with pollutants and microbes. It's not just the public places that are unclean, but we might be amazed at the number of people who do not wash their hands before taking food, after visiting the restroom, or who sneeze without covering their faces. Many infectious diseases such as **typhoid, amoebiasis** and **ascariasis** are transmitted through contaminated food and water.

Advancement in science and technology provide effective controlling measures for many infectious and non-infectious diseases. The use of vaccines and adopted immunization programmes have helped to eradicate **small pox** in India. Moreover a large number of infectious diseases like **polio, diphtheria, pneumonia** and **tetanus** have been controlled by the use of vaccines and by creating awareness among the people.

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